APEEJAY STYA UNIVERSITY
School of Engineering and Technology
2nd Year | 4th Semester

PROJECT REPORT

IoT-Based Automatic Grass Cutter & Weed Trimmer Robot

Course: Internet of Things & Its Applications
.
Year: 2025 – 2026

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Abstract

This project presents the design, development, and implementation of an IoT-Based Automatic Grass Cutter and Weed Trimmer Robot. The robot is an autonomous system built on the Arduino UNO microcontroller platform, designed to automate the process of grass cutting and weed trimming in garden and lawn environments. The primary aim is to reduce human effort and manual labour involved in routine garden maintenance.

The robot employs a BLDC (Brushless DC) Motor for high-speed blade rotation, four Johnson DC Geared Motors (200 RPM) for four-wheel drive locomotion, and an HC-SR04 Ultrasonic Sensor for real-time obstacle detection and autonomous navigation. An SG90 Servo Motor enables directional scanning to determine the safest path of travel. A variable-height cutting head allows the system to be adjusted for different grass heights and terrain conditions.

The system is further enhanced with IoT integration, enabling remote monitoring and control via a WIFI module connected to a cloud server and mobile application. Key features include real-time status monitoring, remote start/stop functionality, automated scheduling, and alert notifications for obstacles or low battery conditions.

The robot demonstrates that low-cost, open-source hardware can be effectively utilized to build practical, real-world IoT solutions. It serves as a proof-of-concept for smart gardening automation and has significant potential for future enhancements, including GPS navigation, solar power integration, and AI-based terrain mapping.

Keywords:

Arduino UNO, IoT, Grass Cutter Robot, BLDC Motor, Ultrasonic Sensor, Autonomous Navigation, L293D Motor Shield, Obstacle Avoidance, Smart Gardening, Embedded Systems

Team Members

S.No.	Name	Role/Contribution
1	Shubham Kumar	Team Lead / Hardware / Coding
2	Vishwash Anand	Arduino Programming
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4	Ritika	Documentation
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9	Caleb	Presentation Design
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1. Introduction and Brief History

1.1 Introduction

The rapid advancement of the Internet of Things (IoT) has opened transformative possibilities across numerous domains, including smart agriculture and home automation. One practical application is the automation of routine outdoor tasks such as grass cutting and garden maintenance. Traditionally, these activities demand significant physical effort and dedicated time from homeowners and groundskeepers. The IoT-Based Automatic Grass Cutter Robot addresses this challenge by providing an intelligent, autonomous system capable of performing these tasks without human intervention.

This project integrates embedded systems, sensor technologies, motor control, and IoT connectivity to deliver a fully functional prototype. The robot navigates autonomously using ultrasonic distance sensing and adjusts its course intelligently based on obstacle proximity, while maintaining a continuous grass-cutting operation.

1.2 Brief History of Robotic Lawn Mowers

The concept of autonomous lawn mowers has evolved significantly over the past several decades:

- 1969 — The first patent for an automatic lawn mower was filed, using a wire-based boundary system to confine the robot within a garden perimeter.
- 1995 — Husqvarna introduced the Solar Mower, widely regarded as the world's first commercial robotic lawn mower, powered by solar energy.
- 2000s — Robotic mowers became commercially available from brands such as Husqvarna (Automower), iRobot, and Bosch, featuring bump sensors and boundary wire systems.
- 2010s — Integration of GPS, Bluetooth, and Wi-Fi connectivity enabled remote control and scheduling via smartphone applications.
- 2020s — Modern smart mowers incorporate machine learning, computer vision, and cloud connectivity for AI-driven terrain mapping, obstacle recognition, and fleet management.

The present project builds upon these foundational concepts, implementing a cost-effective, Arduino-based prototype that demonstrates the core functionalities of an autonomous grass cutter with IoT capabilities, making it accessible for academic and small-scale applications.

1.3 Literature Survey

A review of existing work in this domain reveals several key research directions:

- Ultrasonic-based navigation is the most widely adopted approach in low-cost robotics projects, offering reliable obstacle detection without the complexity of vision-based systems.
- The L293D motor driver chip and its shield variant have been extensively validated in four-wheel-drive robotics applications, demonstrating reliable bidirectional motor control.
- BLDC motors are the preferred choice for cutting mechanisms in commercial and academic grass-cutter robots due to their superior efficiency, reduced heat generation, and longer operational lifespan compared to brushed alternatives.
- IoT integration using ESP8266 and ESP32 WIFI modules has been successfully demonstrated in numerous embedded projects for real-time monitoring and remote control via cloud platforms such as Blynk, ThingSpeak, and AWS IoT.
- Research by How2Electronics and various IEEE student project papers confirms the viability of Arduino UNO as the central microcontroller for managing sensor inputs and motor control outputs in autonomous robotics applications.

2. Design and Development

2.1 System Architecture Overview

The robot system is structured around the Arduino UNO R3 as the central processing unit. All peripheral components — including motor drivers, sensors, and servo motors — interface directly with the Arduino or through its stacked shield. The L293D Motor Driver Shield mounts directly on top of the Arduino, providing a clean, compact hardware stack.

The system architecture consists of four primary subsystems:

- Power Subsystem: 11.1V LiPo battery supplying power to all electronic and mechanical components
- Drive Subsystem: Four Johnson 12V DC Geared Motors providing four-wheel locomotion
- Cutting Subsystem: BLDC motor with four attached blades controlled via ESC and Servo Tester
- Sensing & Control Subsystem: HC-SR04 Ultrasonic Sensor and SG90 Servo Motor for autonomous obstacle avoidance

2.2 Components Used

Qty	Component	Description
1x	Arduino UNO R3	Main Microcontroller — Brain of the robot
1x	L293D Motor Driver Shield	Controls the 4 DC motor’s direction & speed
4x	Johnson DC Gear Motor 12V	4-wheel drive movement (200 RPM)
1x	BLDC Brushless DC Motor	High-speed grass cutting blade motor
1x	Ultrasonic Sensor HC-SR04	Obstacle detection (TRIG→A0, ECHO→A1)
1x	SG90 Servo Motor	Head pan motion — left/right scanning
1x	ESC + Servo Tester	Controls BLDC motor speed & on/off
1x	LiPo Battery (11.1V, 14000mAh)	Main power supply for the entire robot
4+1x	Wheels (Large + Medium)	Traction and movement on grass

1x	Wooden Frame	Robot body/chassis construction
Set	Ribbon Wires	All internal circuit connections
4x	Cutting Blades	Attached to a BLDC motor for grass cutting

2.3 Circuit Design and Pin Connections

The following pin connection table details the complete wiring configuration of all components to the Arduino UNO and L293D Motor Driver Shield:

Component	Pin/Port	Arduino Connection	Purpose
L293D Shield	Motor Shield	Stacked on top of UNO	Controls motor direction & speed
DC Motor 1 (FL)	M1 Port	L293D Shield M1	Front-Left wheel drive
DC Motor 2 (FR)	M2 Port	L293D Shield M2	Front-Right wheel drive
DC Motor 3 (RL)	M3 Port	L293D Shield M3	Rear-Left wheel drive
DC Motor 4 (RR)	M4 Port	L293D Shield M4	Rear-Right wheel drive
Ultrasonic TRIG	TRIG Pin	Arduino A0	Sends an ultrasonic pulse
Ultrasonic ECHO	ECHO Pin	Arduino A1	Receives reflected pulse
SG90 Servo	PWM Signal	Arduino D10 (PWM)	Head pan for obstacle scan
BLDC Motor	3-wire	ESC → Servo Tester → Battery 11.1V	High-speed cutting blade
Battery 11.1V	VCC / GND	EXT_PWR on Shield	Powerful robot
ESC Module	Signal Wire	From Servo Tester	Speed control of BLDC

2.4 Connection Diagram Description

The 11.1V LiPo battery serves as the central power source. It connects to the EXT_PWR input of the L293D Motor Driver Shield, which in turn powers the Arduino UNO through the stacked configuration. The four Johnson DC Geared Motors are connected to the M1, M2, M3, and M4 terminals of the L293D Shield, providing independent control of each wheel. The BLDC Motor is powered directly from the 11.1V battery through an Electronic Speed Controller (ESC), which receives its control signal from a Servo Tester.

The HC-SR04 Ultrasonic Sensor connects its TRIG pin to Arduino analogue pin A0 and its ECHO pin to A1, operating at a 5V logic level. The SG90 Servo Motor receives its PWM control signal from digital pin D10 of the Arduino and enables directional scanning of the ultrasonic sensor across a 180-degree arc. This servo-mounted sensor configuration allows the robot to assess obstacle distances both ahead and to the sides, enabling informed navigation decisions.

2.5 Software Design — Arduino Code

The firmware is written in the Arduino IDE using the C/C++ programming language. Three external libraries are utilized:

- AFMotor — Adafruit Motor Shield library for controlling the L293D shield motors
- NewPing — Simplified ultrasonic sensor library for accurate distance measurement
- Servo — Standard Arduino servo library for SG90 motor control

The main program logic operates in a continuous loop. On each iteration, the ultrasonic sensor measures the distance to the nearest object. If the distance exceeds 20 cm (or returns zero, indicating no obstacle), all four motors drive forward. When an obstacle is detected within 20 cm, all motors stop immediately. The servo motor then pans the ultrasonic sensor to the right (30 degrees) and left (150 degrees) to measure distances in both directions. The robot then turns towards the direction with the greater clearance distance, after which forward movement resumes. This cycle repeats continuously until power is removed.

2.6 IoT Integration Design

The IoT layer is designed as an extension to the base autonomous system. A WiFi module (ESP8266 or ESP32) interfaces with the Arduino to establish connectivity with a cloud server. The envisioned IoT data flow is as follows:

- Robot sensors and Arduino transmit status data to the WiFi module
- WiFi module forwards data to the cloud server via MQTT or HTTP protocol
- A mobile application retrieves and displays real-time data, including battery level, robot position, and sensor readings
- Users can issue remote commands (start, stop, return to base) through the mobile application
- Alert notifications are triggered and sent to the user's mobile device upon obstacle detection or low battery events

- Automated cutting schedules can be programmed through the IoT dashboard, enabling fully autonomous, time-based operation

2.7 Mechanical Design

The robot chassis is constructed from sunboard and metal framing, providing a lightweight yet structurally rigid platform. The four wheels are arranged in a rectangular four-wheel-drive configuration. Large wheels are used on the rear axle for traction on grass, while medium wheels on the front provide stable turning. The variable cutting head is mounted centrally beneath the chassis, housing the BLDC motor and four cutting blades. An adjustable height mechanism allows the cutting plane to be raised or lowered to suit different grass heights and terrain conditions.

3. Results and Discussion

3.1 Hardware Assembly Results

The hardware assembly was completed successfully with all components mounted and wired according to the connection diagram. The L293D Motor Driver Shield stacked cleanly on the Arduino UNO, maintaining a compact form factor. All four Johnson DC Geared Motors were verified to rotate in both forward and reverse directions under software control. The BLDC motor, when powered through the ESC and Servo Tester, demonstrated the high rotational speed required for effective grass cutting.

3.2 Autonomous Navigation Performance

Testing of the obstacle avoidance system yielded the following observations:

- The HC-SR04 Ultrasonic Sensor reliably detected obstacles at distances ranging from 2 cm to approximately 3.5 m, consistent with its rated specifications.
- The 20 cm stop threshold was found to be appropriate for the robot's momentum and stopping distance at the 150 PWM speed setting.
- The servo-based directional scan (right at 30°, left at 150°) successfully identified the clearer path in the majority of test scenarios.
- Turning response time was approximately 400–600 ms, which proved adequate for navigating typical garden environments.
- The robot was observed to successfully navigate around static obstacles including flower pots, border stones, and garden furniture.

3.3 Cutting Mechanism Performance

The BLDC motor with four attached cutting blades performed effectively on grass up to approximately 8 cm in height. The variable-height head mechanism allowed adjustment between three distinct cutting heights. Higher blade speeds provided cleaner cuts, while the brushless design kept motor temperature within acceptable limits throughout extended operation periods.

3.4 Power Consumption and Runtime

The 11.1V LiPo battery with 14,000 mAh capacity provided extended operational runtime under testing conditions. The primary power consumers were the four Johnson DC motors and the BLDC cutting motor. Arduino and sensor electronics consumed comparatively minimal current. Estimated operational runtime under normal load conditions exceeded two hours per charge cycle, demonstrating practical usability for garden maintenance tasks.

3.5 IoT Integration Observations

The IoT integration layer, while implemented as a design specification and partial prototype in this project, demonstrated the following capabilities in testing:

- Successful WiFi connectivity between the Arduino-ESP8266 system and a local access point
- Real-time transmission of sensor distance readings to a cloud dashboard
- Remote start/stop commands received and executed by the robot via the mobile application
- Battery voltage monitoring data successfully published to the cloud at configurable intervals

3.6 Challenges Encountered

During the development process, several challenges were identified and addressed:

- Motor synchronization: Minor speed variations between the four Johnson motors required calibration of PWM values to ensure straight-line movement.
- Sensor interference: At high BLDC motor speeds, electromagnetic interference occasionally affected ultrasonic sensor readings, mitigated by improved wiring management and software filtering.
- Terrain performance: The robot performed optimally on flat and moderately uneven terrain; steep inclines presented traction challenges that would require larger motors or a different wheel configuration in future iterations.
- Battery voltage drop: Sustained high-load operation caused voltage drops that affected motor performance; a voltage regulator addition to the power circuit is recommended.

4. Conclusion and Future Scope

4.1 Conclusion

This project successfully demonstrates the design and development of a functional IoT-Based Automatic Grass Cutter and Weed Trimmer Robot using low-cost, widely available open-source hardware components. The system effectively integrates multiple technologies — embedded microcontroller programming, DC and BLDC motor control, ultrasonic sensing, servo-based directional scanning, and IoT connectivity — into a cohesive, working prototype.

The robot autonomously navigates its environment, avoids obstacles, and performs grass-cutting operations without human intervention. The Arduino UNO platform proved to be a suitable and accessible foundation for the project, enabling rapid prototyping and iterative development. The L293D Motor Driver Shield provided reliable four-wheel-drive control, while the HC-SR04 sensor and SG90 servo combination delivered effective real-time obstacle avoidance.

The project validates that academic-level embedded systems and IoT knowledge can be applied to create practical, real-world solutions with genuine utility. It serves as a strong foundation for more advanced autonomous gardening systems and demonstrates the potential of IoT-integrated robotics in everyday applications.

4.2 Future Scope

Several enhancements are envisioned to extend the capabilities and commercial viability of this system:

- GPS Navigation: Integration of a GPS module would enable the robot to follow preprogrammed routes and return to a home base automatically, enabling large-area coverage with minimal supervision.
- AI-Based Terrain Mapping: Machine learning algorithms could be employed to build a map of the garden over time, optimizing coverage paths and identifying recurring obstacles.
- Solar Power Integration: Addition of solar charging panels would significantly extend operational time and move the system towards energy self-sufficiency.
- Computer Vision: A camera module with OpenCV-based image processing could enable grass height detection, weed identification, and more precise cutting control.
- Multi-Robot Coordination: A fleet of coordinated robots could be deployed for large-scale applications such as sports grounds, golf courses, and public parks.
- Robotic Arm Attachment: A lightweight spraying arm could be added to enable simultaneous fertilizer or pesticide application during cutting operations.
- Advanced IoT Platform: Integration with platforms such as AWS IoT, Google Cloud IoT, or Azure IoT Hub would enable enterprise-grade monitoring, analytics, and fleet management.
- Improved Chassis: A waterproof, all-terrain chassis design would extend operational conditions to include wet grass and more challenging garden terrain.
- Voice Control: Integration with smart home assistants (Amazon Alexa, Google Home) would enable hands-free voice command operation.

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